

MA 301
Test 2
Fall - 2014

Problem	points	out of
1		12
2		5
3		5
4		5
5		5
Score		32

Name: _____

HR: _____

Signature: _____

Problem 1: Let $z_1 = -3i$, $z_2 = 3 - 3i$, $z_3 = -5$, $z_4 = -\sqrt{3} - i$.

(a) Perform the following operations and leave answer in standard form. (No Decimals)

▷ $z_1 - z_4$

▷ $\frac{z_4}{z_2}$

(b) Write z_1, z_2, z_3, z_4 in Euler's polar form. (No Decimals)

▷ $z_1 =$

▷ $z_2 =$

▷ $z_3 =$

▷ $z_4 =$

(c) Perform the following operations and leave answer in Euler's polar form. (No Decimals)

▷ $z_1 z_2$

▷ $(z_1 z_2)^6$

(d) Solve the complex equations for x and leave your solutions in Euler's Form: $re^{\theta i}$. (No Decimals)

▷ $\frac{x^{1/6}}{z_1} - z_2 = 0.$

▷ $x^3 - z_3 = 0.$

Problem 2: Let $\mathbf{F} = [-3xz^2, 6, 5x^2y]$ and S is the parabolic cylinder $S : y = x^2$ where $0 \leq x \leq 2$ and $0 \leq z \leq 3$ and normal vector point away from the y -axis. Compute the flux of $\mathbf{F}$ through the parabolic cylinder S

Problem 3: Let $\mathbf{F}$ and S be the same as in **Problem 2**. If C be the boundary of S . Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

Problem 4: Let $\mathbf{F} = [x, ye^z, e^z]$ and S the surface (outward orientation) of the box $|x| \leq 1, |y| \leq 2, -1 \leq z \leq 7$. Compute the flux through S .

Problem 5: Let $F = [x^3 - y^3, y^3 - z^3, z^3 - x^3]$ and S is the surface of the solid bounded by:

- (i) quarter sphere $x^2 + y^2 + z^2 = 25$ ($y \geq 0, z \geq 0$);
- (ii) the plane $z = 0$;
- (iii) the plane $y = 0$

with outward pointing normal. Compute the flux through S .